

R4Tun:LLM-guided adaptive segmental tunnel lining segmentation in point clouds

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We are grateful to the Associate editor and reviewers for their time in reviewing the manuscript and for their useful and insightful feedback. We have responded to the queries individually below and modified the manuscript as appropriate.

Associate Editor

- Avoid orphan headings without context: these are headers that are immediately followed by the next header, without any text in between (line 85-86, 169-170, ..). At least add a brief introduction to explain what will be covered in the section.

Short introductory paragraphs have been added at the opening of Related work, Methodology, Results, and Discussion.

- fix reference on line 191. Also in other places, the reference to the Appendix is missing. All references in the PDF need to resolve to a number. The Appendix can be directly included in the manuscript after the references.

Unresolved appendix references have been corrected. All appendix references now use numbers. The Appendix is directly included in the manuscript.

- Limitations must be included and discussed in the conclusion. Limitations are also not necessarily the same thing as Future Work, as some limitations are simply inherently part of the proposed approach (and thus need to be highlighted in a conclusion as well).

The Conclusion now discusses the main limitation, including:

- evaluation on SAM4Tun and Seg2Tunnel only;
- single expert-tuned reference anchor;
- inherited SAM4Tun operator assumptions and failure modes;
- dependence on offline bounds and knowledge quality;
- single-pass adaptation without iterative self-correction.

- Do a thorough check of all references such that they follow the author guidelines and include full bibliographic details. Author names, journal or book titles, chapter or article titles, year of publication, volume numbers, article numbers or pagination must be included, where applicable. Several names are mistyped, likely due to improper handling

in LaTeX. Include DOIs for articles and ISBNs for book volumes. Check whether arXiv preprint articles have not been published by now, and try to replace them in that case, such that you can rely more on peer-reviewed and validated sources rather than pre-prints. Be careful with capitalization in titles as well, when using LaTeX; many acronyms are now in lower-case. When citing an article with 6+ authors, only the first six authors need to be listed, and the remaining ones can be replaced by et al. More guidance on this can be found in the author guide of the journal, and the examples it refers to.

The reference list has been checked for author names, article titles, journal titles, volume/article numbers, page ranges, DOIs, capitalisation, and published versions of preprints where available. LaTeX capitalisation errors for acronyms have also been corrected.

Reviewer 1

1. The manuscript is generally well revised and significantly improved. One minor suggestion is to further strengthen the discussion of recent studies on intelligent tunnel inspection, adaptive parameter tuning, and AI-assisted point-cloud analysis for underground infrastructure. In particular, the authors may consider citing several recent related works to better position the contribution of R4Tun within the latest development of LLM-/AI-enabled tunnel analysis and infrastructure inspection research:

DOI: 10.1016/j.aei.2026.104774

DOI: 10.1016/j.autcon.2026.106769

DOI: 10.1016/j.tust.2024.105998

DOI: 10.1016/j.measurement.2023.113903

We thank the reviewer for this positive assessment. The Related Work and Discussion sections have been updated to include recent work on intelligent tunnel inspection, adaptive parameter tuning, and AI-assisted point-cloud analysis.

2. The proposed framework currently evaluates adaptation quality mainly through segmentation metrics (mIoU and OA). It would be helpful if the authors could briefly discuss whether the adapted parameters remain stable across repeated runs or different API calls, especially considering that LLM-based reasoning may still introduce slight stochasticity even under low-temperature settings. A short discussion on reproducibility would further strengthen the practical applicability of the framework.

We added a repeatability analysis (Appendix / Table X). On 30 tunnels, each LLM was re-inferred twice under m+s+k with temperature 0. On average,

- about mean 90.9% of the 18 critical parameters were unchanged between run 1 and 2.
- mean $|\Delta mIoU|$ between the two runs was 0.029 (median 0.000).

3. The paper would benefit from a short clarification regarding the computational overhead introduced by the multi-agent adaptation process. Although runtime and API costs are reported, a brief comparison between the adaptation time and the original SAM4Tun inference time would help readers better understand the trade-off between adaptability and efficiency in practical deployment scenarios.

We have added a direct runtime comparison in Section 4.5. We separated the base SAM4Tun processing time (235 s) from the additional LLM adaptation time.

LLM	Extra time	Total time	$\Delta mIoU$	mIoU gain per +100 s overhead
Gemini-3-Flash	+96 s (0.4×)	~331 s	+0.166	~0.173
Opus-4.6	+140 s (0.6×)	~375 s	+0.192	~0.137
GPT-5.4	+307 s (1.3×)	~542 s	+0.171	~0.056

4. In Section 5.2, the limitations are discussed clearly. As a minor improvement, the authors could also briefly comment on how the proposed framework might generalize to other point-cloud processing pipelines beyond SAM4Tun, for example whether the context-design strategy (memory-state-knowledge) is expected to remain transferable even when the downstream segmentation backbone changes.

The Discussion (Section 5.2) acknowledges that transferability is conceptualised without empirical validation. The m+s+k design can apply to other parameter-sensitive multi-stage point-cloud pipelines if four conditions hold:

- bounded tunable stage parameters;
- reference configurations with comparable target characterisation;
- summarisable intermediate state;
- and encodable stage-specific knowledge.

Reviewer 2

I would like to commend the authors for their efforts in revising this manuscript. The authors have clearly taken the previous reviewers' comments. The expansion of the dataset to 30 tunnel subsets, the introduction of a deterministic non-LLM rule-based baseline, the rigorous cumulative ablation studies, and the inclusion of comprehensive statistical analyses have significantly elevated the methodological rigor of the paper. The concept of utilizing LLMs as white-box reasoning agents to adapt expert-designed pipelines via structured context is interesting.

We thank the reviewer for this positive assessment and for highlighting the concept of our approach. We have clarified the focus of our study within the Abstract, Introduction, and Conclusion of the revised manuscript.

However, despite the exhaustive experimental design, I have a fundamental and critical concern regarding the performance of the proposed system. The empirical results, as currently presented, fall drastically short of the standards required for practical deployment in construction automation, which severely undermines the core premise of the paper.

We agree. We have clarified that this study aims to explore a controlled, cross-LLM, zero-label, no-retraining, and no-expert-tuned self-adaptation mechanism, rather than to compete directly with existing segmentation benchmarks. By operating in a single pass (~331 s to ~542 s), this framework introduces explicit LLM reasoning into an expert-designed pipeline, serving as a methodological foundation for future development.

Massive Misclassification: An OA of ~53% implies that nearly half of the point cloud is incorrectly classified. An mIoU of ~33% indicates a drastic lack of overlap between the predicted segment masks and the ground truth.

We agree. Our error analysis identifies the primary performance bottlenecks as SAM4Tun's fixed configuration, particularly regarding A/B offset calculations and segment direction. We plan to address these limitations in future iterations by loosening these constraints, thereby enabling reasoning models to fully leverage their inherent self-adaptation capabilities.

Failure in Downstream Tasks: In the context of automated tunnel inspection (e.g., structural health monitoring, deformation analysis, water leakage detection, or Scan-to-BIM workflows), a segmentation pipeline with a ~50% error rate is practically unusable. The downstream geometric and structural analyses would be entirely compromised by the massive amount of noise, missing segments, and misidentified structural components.

We agree. This study is intended as an initial exploration of the self-adaptation capabilities of reasoning models rather than a ready-to-deploy tool for downstream applications. Accordingly, we have clarified in the Introduction, Discussion, and Conclusion that the current absolute accuracy does not yet support autonomous final inspection.

The Low Baseline Fallacy: The authors celebrate a >100% relative improvement, but this is a classic statistical illusion caused by a catastrophically failing baseline. The static SAM4Tun baseline completely collapses on off-reference tunnels (overall mIoU = 0.15; complex tunnels = 0.04). Recovering from a state of total failure to a state of mediocre/poor performance (mIoU 0.34) does not validate the framework for real-world engineering applications.

We agree. We have removed all relative improvement metrics and now focus exclusively on absolute performance metrics for discussion throughout the manuscript.

Poor Performance on Regular Tunnels: Most concerningly, even on the regular tunnel subsets—which closely resemble the expert-tuned reference configuration—the mIoU only reaches ~0.50-0.54. If an LLM, armed with explicit memory, intermediate state, and expert knowledge documents, cannot achieve a high degree of accuracy (e.g., mIoU > 0.80) on in-distribution or near-reference data, it raises serious doubts about either the LLM's actual reasoning capability in this domain or the fundamental brittleness of the underlying SAM4Tun pipeline.

We agree. Our error analysis identifies the primary performance bottlenecks as SAM4Tun's fixed configurations and heavily relying on K position.

Q1. Justification of Practical Applicability: The primary scope of this journal is automation in construction. How do the authors justify the practical deployment of a system that misclassifies roughly 50% of the structural point cloud? The authors must provide a realistic discussion on what specific, limited engineering tasks (if any) an mIoU of 0.34 can actually support, rather than broadly claiming it "reduces manual per-tunnel re-tuning."

We agree. We have removed all claims of "reduces manual per-tunnel re-tuning." through the new manuscript and clarify that the current absolute accuracy does not yet support autonomous final inspection.

Q2. In-depth Error Analysis: The manuscript currently lacks a qualitative and quantitative error analysis. Where are these massive errors concentrated? Are the LLMs failing to identify the correct boundaries? Is the 2D-to-3D reprojection introducing severe artifacts? Are background noises being classified as key blocks? The authors must

provide visual error maps (e.g., False Positives vs. False Negatives) and analyze why the absolute metrics remain so stubbornly low despite LLM intervention.

We agree. A new error-analysis subsection, Section 4.5 (Error analysis), has been added to report false-positive/false-negative error maps (Fig. error map) and a detailed failure decomposition on representative regular (tunnel 1-1) and complex (tunnel 4-1) examples. Section 4.5 further analyses four structural limits of the fixed SAM4Tun labelling substrate (Fig. error mechanism): ring-level non-uniform point density, a moving K anchor, fixed segment-offset template drift, and hard-coded walk direction. The Discussion (Section 5.1) now links these failure modes to the stated absolute performance ceiling and to the mechanism-level interpretation of the adaptation results.

Q3. Comparison with State-of-the-Art (SOTA) Baselines: The current comparison is strictly internal (LLM vs. Non-LLM vs. Static SAM4Tun). How does an absolute mIoU of 0.34 compare to existing, traditional geometric feature-engineering methods or standard supervised 3D deep learning models trained on similar datasets? If a standard, non-LLM baseline can achieve an mIoU of 0.65+, the justification for introducing the computational overhead and API costs of LLMs is severely weakened.

We agree. rather than to compete directly with existing segmentation benchmarks.

Q4. The Ceiling of the Pipeline: The authors briefly mention the "single-reference design" as a limitation. However, if the underlying SAM4Tun pipeline is inherently incapable of achieving high mIoU due to its 3D-to-2D projection and SAM-prompting mechanics, then using an LLM to tune its parameters is merely "rearranging deck chairs on the Titanic." The authors need to clarify whether the low mIoU is a failure of the LLM's parameter selection or a fundamental limitation of the SAM4Tun architecture itself.

We agree. We have identified four structural constraints within the fixed SAM4Tun labeling substrate that contribute to these performance bottlenecks:

1. **Non-uniform point density:** Central rings exhibit significantly higher point density than end rings. A single tunnel-level preprocessing setup cannot accommodate both, leading to failures in ring-boundary and K-anchor detection on sparser rings.
2. **Moving K-anchor:** Staggered ring assembly shifts the K-anchor position along the arc between rings. If the K-anchor is misidentified, the entire ring's label projection rotates, resulting in bimodal per-ring accuracy.
3. **Fixed segment-offset template:** Each block is labeled relative to K with a fixed angular offset. Cumulative sector-size mismatches increase with distance from K, causing recall to be lowest for segments furthest from the anchor.

4. **Hard-coded walk direction:** B- and A-blocks are assigned on a fixed side of K. If a ring's physical handedness is reversed, labels become a mirror image of the ground truth, which parameter tuning alone cannot correct.

However, these structural constraints do not invalidate the SAM4Tun pipeline; instead, they define the exact configuration challenges that the reasoning model's self-adaptation capacity is designed to address. As outlined in our future work, leveraging this capacity to evolve the pipeline beyond rigid templates toward dynamic, context-aware parameterisation will enable effective adaptation to these structural variations.

The integration of LLMs for context-aware parameter adaptation is interesting. The authors have done a good job structuring the experiments and ablation studies. Unfortunately, the actual numerical outcomes (mIoU ~0.33, OA ~0.53) reveal that the resulting system is riddled with segmentation errors and is currently unfit for practical, real-world tunnel inspection. Unless the authors can provide a compelling, evidence-based justification for why these low absolute metrics are acceptable, or demonstrate that the errors are benign for specific downstream tasks, the manuscript in its current form does not meet the practical impact threshold expected by the readership of Automation in Construction.

We agree. To address the reviewers' concerns, we have revised the manuscript to better frame the study's contribution and clarify the performance limitations. Here is a summary of the actions taken:

- **Reframed Study Scope:** We explicitly repositioned the manuscript as a "mechanism-contribution" study rather than a performance-competitive segmentation method. We clarified that R4Tun explores a zero-label, no-retraining, no-expert-tuned self-adaptation mechanism and does not yet support autonomous final inspection.
- **Correction of Performance Metrics:** We removed all relative improvement metrics to avoid misleading comparisons. The manuscript now focuses exclusively on absolute performance (mIoU and OA) and acknowledges the current performance ceiling.
- **In-Depth Error Analysis:** We added Section 4.5 (Error Analysis), which includes visual error maps (False Positives vs. False Negatives) and identifies four specific structural bottlenecks (non-uniform point density, moving K-anchor, fixed segment-offset template drift, and hard-coded walk direction).
- **Benchmarking and Comparisons:** We included a new comparison table (Table X) situating R4Tun against established supervised and foundation-model tunnel point-cloud segmentation methods for context.

- **Clarification of LLM Role:** We detailed the agent design (Memory-State-Knowledge) and the five-step Chain-of-Thought (CoT) protocol in Section 3.4, and added a deterministic rule-based baseline comparison (Section 3.5.2) to distinguish LLM reasoning from rule-based adjustment.
- **Limitations & Future Work:** We updated the Conclusion to explicitly include limitations (e.g., dependence on offline bounds, single-pass adaptation) and linked our identified structural bottlenecks directly to the future development of dynamic, context-aware parameterization models.

Reviewer 3

The study objectives and general rationale are mostly clear. However, the role and generality of the LLM in the proposed framework require more precise framing. The current wording may suggest that R4Tun provides a general LLM-based tunnel segmentation capability, whereas the implemented workflow still relies on an expert-designed pipeline, including tunnel unfolding, denoising, image enhancement, SAM-based segmentation, and 3D reprojection. Please revise the motivation, especially around Fig. 1, to clarify that the main contribution is LLM-guided bounded parameter adaptation for an existing segmentation pipeline, rather than a generally capable LLM solution for tunnel segmentation.

We agree. We have clarified that this workflow still relies on an expert-designed pipeline rather than providing a general LLM-based tunnel segmentation capability in Fig. 1 and throughout the revised manuscript.

The method is generally described in sufficient detail, but some clarification would improve reader understanding. Additional explanation is needed to clarify how the LLM converts tunnel characteristics, reference parameters, intermediate pipeline states, and stage-specific knowledge documents into meaningful bounded parameter updates. Without this explanation, the specific role of LLM-based reasoning remains difficult to distinguish from deterministic rule-based parameter adjustment.

We agree. Section 3.4 (Agent design, Context design, CoT design) and Algorithm 1 describe how each stage agent assembles memory, state, and knowledge into a prompt and runs a five-step CoT protocol. Section 3.5.2 and the rules appendix define the deterministic Non-LLM baseline (same knowledge as explicit if-then lookups on characterisation fields).

Section 4.1, Table 4, and Fig. 9 compare Non-LLM vs LLM: on regular tunnels, rules do not beat the static baseline; LLM (m+s+k) does; m+s exceeds the rule baseline by 0.10–0.12 mIoU overall.

Appendix D gives the denoising agent's memory, state, and knowledge inputs;
Appendix E gives a full five-step CoT trace and returns JSON for one tunnel.

The manuscript would benefit from improving several existing figures and tables so that they are more self-contained.

- Fig. 1: The motivation in Fig. 1 should be revised to avoid overstating the role of the LLM. The figure should clarify that R4Tun reduces repeated deployment-stage expert retuning by reusing expert-defined reference parameters, parameter bounds, and knowledge documents, rather than removing expert knowledge from the workflow.

Fig. 1 has been revised to clarify that R4Tun reuses expert-defined reference parameters, parameter bounds, and knowledge documents. The figure now emphasizes reduced deployment-stage retuning within a predefined pipeline, rather than removal of expert knowledge.

- Fig. 2: Some labels, such as B1 and K2, are difficult to understand from the figure alone. These labels should be defined directly in the figure or caption so that readers do not need to refer back to the main text.

We have revised Fig. 2 and its caption to define these labels directly.

- Fig. 5: The relationship between the three interpolation methods shown on the left and the workflow shown on the right is ambiguous. The figure should clarify whether these methods are used sequentially or presented as alternatives.

We have revised Fig. 5 and its caption to clarify that the interpolation methods are sequential steps in one workflow.

- Table 7: Some parameters, especially the underlined parameters, are not sufficiently explained. The authors should add short descriptions of what these parameters control and how they affect the corresponding stages of the segmentation pipeline.

We have added short descriptions for the underlined parameters in Table 7. The revised table now explains what each parameter controls and how it affects the corresponding stage of the segmentation pipeline.

The strengths of the study are present, but they could be emphasised with greater precision. The manuscript should more clearly state that the strength of R4Tun lies in

integrating feature-engineered preprocessing, a foundation segmentation model, and LLM-guided bounded parameter adaptation. This framing would better highlight the actual contribution and avoid overstating the LLM as a general-purpose tunnel segmentation model.

We have revised the Abstract, Introduction, and Conclusion to make the scope of the framework clearer.

The limitations are generally stated, but the manuscript should more explicitly acknowledge that R4Tun does not eliminate the need for expert knowledge. Expert input is still required to define reference parameters, parameter bounds, and stage-specific knowledge documents.

We have made this limitation explicit in the Discussion and Conclusion. The revised manuscript presents the LLM as adapting within these expert-defined constraints, not replacing them.

The manuscript structure is generally clear, but the framing and figure explanations require revision to improve flow.

We have revised the Abstract, Introduction, and Conclusion to reframe the scope of the framework. We have also revised the captions of Figs. 1, 2, and 5 to improve readability and manuscript flow.

We hope these revisions address all reviewer concerns.